Triwizathon Problem Statements:

Problem Statement 1: Retail Expansion Intelligence Platform

Choosing the right location for a new retail store is a complex decision that requires analysing multiple factors. A location may have a large population but intense competition, high rental costs, or limited accessibility. Relying on manual research or individual judgement can lead to poor investment decisions and missed business opportunities.

Your challenge is to develop **an AI-powered Retail Expansion Intelligence Platform** that helps retail chains identify and evaluate the most promising locations for their next store.

The solution should use Artificial Intelligence and data-driven analysis to evaluate geographical and demographic information, competitor locations, existing store coverage, accessibility, and estimated operating costs. AI should help identify patterns, estimate potential customer demand, predict the commercial viability of candidate locations, and rank them according to their suitability for expansion.

The platform should also provide AI-generated, evidence-based explanations for its recommendations, highlighting the factors that make a location promising, potential risks, and how it compares with alternative locations. Teams may incorporate machine learning, predictive modelling, geospatial AI, or other suitable AI techniques to support their decisions.

Expected Outcome: A functional AI-powered software solution that analyses potential retail locations, predicts their business potential, ranks them using data-driven methods, and provides explainable recommendations to support retail expansion decisions.

Note: The use of AI is mandatory and must contribute meaningfully to the analysis, prediction, ranking, or recommendation process. Simply integrating a chatbot or using an AI API to generate text without performing meaningful analysis will not satisfy this requirement. Teams may use publicly available geographical data or create realistic simulated datasets. All assumptions, data sources, and limitations should be clearly stated.

Problem Statement 2: Distribution Intelligence & Inventory Optimization Engine

FMCG companies distribute products across multiple warehouses, distributors, and retail stores. However, differences in regional demand, inventory availability, transportation costs, and product shelf life make efficient distribution challenging. Some locations experience stockouts while others hold excess inventory, resulting in lost sales, increased costs, and product wastage.

Your challenge is to develop a **Ai Powered Distribution Intelligence Engine that helps** companies make smarter product distribution and inventory allocation decisions.

The solution should analyse available inventory, estimated demand, and distribution constraints to recommend which products should be sent to which locations, in what quantities, and from which warehouses. It should aim to minimise stockouts, unnecessary transportation costs, and inventory wastage while making the best possible use of available stock.

Teams are encouraged to incorporate demand forecasting, inventory optimisation, or other intelligent decision-making techniques to improve their recommendations.

Expected Outcome: A functional software solution that generates actionable product distribution recommendations and demonstrates how its decisions can improve supply-chain efficiency.

Note: Teams may use publicly available data or create realistic simulated datasets. All assumptions should be clearly stated.

Problem Statement 3: Prompt Optimization & Token Efficiency Engine

Writing effective prompts for AI models is essential for getting accurate results, but poorly structured prompts often lead to unnecessary token wastage and high API costs. A prompt may be overly verbose, repetitive, or lack clear constraints, causing models to consume excessive computational resources without improving output quality. Conversely, cutting down a prompt too aggressively can lead to incomplete or inaccurate answers.

Your challenge is to **develop a Prompt Optimization and Token-Efficiency Engine** that evaluates a user's raw prompt and generates two optimized alternatives: a high-context version and a low-token version.

The solution should analyze user intent, structure, and token consumption to provide two specific variations. The first should be a comprehensive, context-rich prompt tailored for complex reasoning and high-precision tasks. The second should be an ultra-lean, token-efficient prompt engineered to deliver the desired result using minimal tokens to prevent wastage and reduce inference costs.

The platform should provide a clear comparison between both versions, highlighting the estimated token count, expected trade-offs, and recommendations on when to use each variant.

Expected Outcome: A functional software solution that analyzes input prompts, generates both high-context and low-token alternatives, and provides token usage comparisons to help users balance performance with cost efficiency.

Note: Teams may use standard tokenizer libraries or LLM APIs to calculate token counts and demonstrate prompt effectiveness. All assumptions should be clearly stated.

Problem Statement 4: Esports Scouting & Multi-Roster Intelligence Platform

Building and managing competitive esports teams is a high-stakes challenge for modern organizations. Scouting decisions are often driven by subjective opinions or surface-level stats rather than holistic performance data, leading to poor team chemistry, wasted transfer budgets, and sub-optimal roster synergy. Furthermore, managing multiple rosters across completely different titles simultaneously makes it difficult to maintain structured talent pipelines and track player development at scale.

Your challenge is to **develop an AI-Powered Esports Scouting** and Multi-Roster Intelligence Platform that helps esports organizations scout, evaluate, and recruit the right players while centralizing roster operations.

The solution should ingest and analyze player match data, situational metrics, and role-specific performance indicators—such as shot-calling impact and mid-round decision-making for In-Game Leaders (IGLs)—to rank and recommend candidate profiles from a talent pool. Additionally, the platform should support multi-game roster management, allowing organizations to monitor team composition, track performance trajectories across different game ecosystems, and assess potential squad synergy.

The platform should provide data-backed talent recommendations with role-specific insights, justifying why a particular player is the ideal fit for a roster's strategic needs over the alternatives.

Expected Outcome: A functional software solution that analyzes player data pools, identifies and ranks ideal candidates for specific competitive roles (such as IGLs), and provides unified roster analytics and AI-driven scouting recommendations across multiple titles.

Note: Teams may use public competitive APIs, community esports datasets, or realistic simulated match logs. All metrics, role evaluation formulas, and assumptions should be clearly stated.

Problem Statement 5: Grocery Intelligence & Food Waste Management System

Households purchase groceries regularly, yet managing these purchases efficiently remains a challenge. Products often get forgotten at the back of refrigerators or storage shelves, expiry dates go unnoticed, duplicate purchases accumulate, and food is wasted even when it could have been consumed on time. At the same time, consumers may struggle to maintain an accurate record of their grocery inventory, understand their purchasing patterns, and make informed decisions about what to buy, consume, or replenish.

The challenge becomes more complex when grocery information is scattered across physical packaging, purchase receipts, shopping applications, and individual memory. Retailers also maintain product and billing information within their own systems, creating potential opportunities for better coordination between grocery purchases and household inventory management.

Your challenge is to develop **an Ai Powered Intelligent Grocery and Food Waste Management System** that explores how household grocery data can be used to improve inventory visibility, purchasing decisions, product awareness, and food waste reduction.

Expected Outcome: A functional software solution that demonstrates an innovative approach to household grocery intelligence, helping users understand their inventory, make informed purchasing and consumption decisions, and identify opportunities to reduce avoidable food waste. The solution should demonstrate its effectiveness using realistic sample data or publicly available datasets, with assumptions and limitations clearly documented.

Note: Teams may use publicly available datasets, synthetic grocery inventories, simulated purchase receipts, and realistic household consumption scenarios. Any product, nutritional, or expiry information used must be handled responsibly and validated against reliable sources where possible.